



ASPLoC Project (Revised 2025)

Shiming Li, Rashid Aligholipour, Yuan Yao



UPPSALA
UNIVERSITET

What you will do

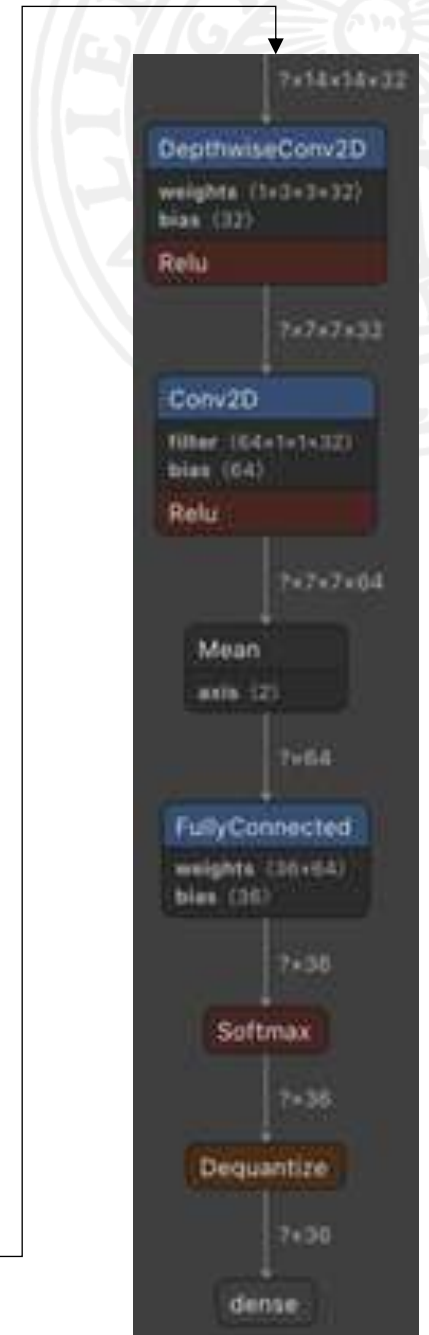
- **Deploy a neural network on a development board**
 - Image classification of handwritten digits and letters
 - Only inference (forward-propagation), no learning needed
- **Analyze the program behavior:**
 - What does each layer do?
 - How much time does each step take?
- **Use the FPGA on the board to accelerate matrix multiplication**
 - Make the ARM processor and the FPGA work together



What you will do

- **The neural network:**

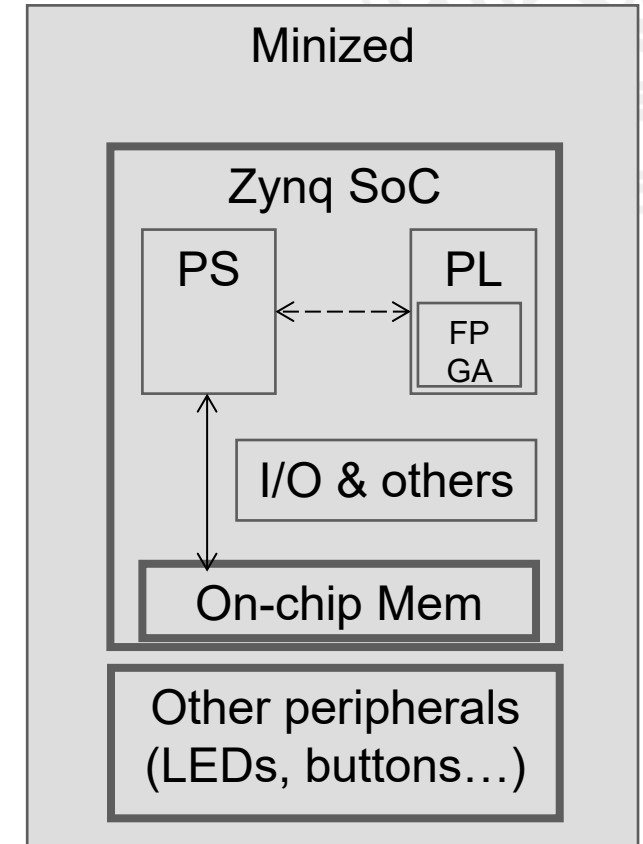
- A CNN, with conv+depthconv structure
- You are given both the .tflite format (from STM32 Model Zoo) and the C code (generated with a modified version of TinyEngine)
- Trained on Extended MNIST--handwritten letters and numbers
- Input 28*28, output 36 neurons
- Find the st_emnist network on Studium! You can test it out (it's C) on your own machine
- Later, add the files into your Vivado Vitis project to use the code on the board's processor



What you will do

- **The board:**

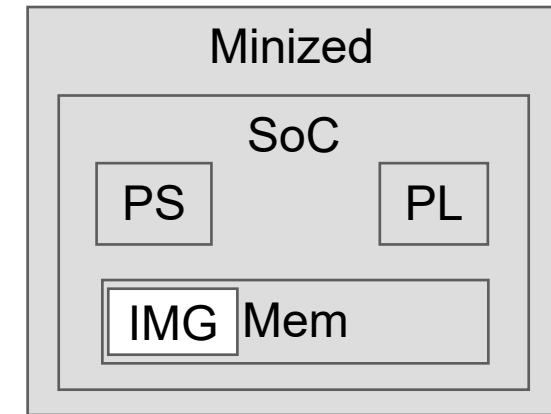
- Avnet Minized-7Z007-G (With Zynq-7000 SoC)
- Processing system (PS): an ARM Cortex-A core
- Programmable logic (PL): contains a piece of FPGA
- The PS can control the PL via AXI interface
- The PS can allow the PL to access the mem via DMA
- Follow the tutorials on Studium for some hands-on experience!
- We will have a lecture reflecting on the tutorials



What you will do

- **The task:**

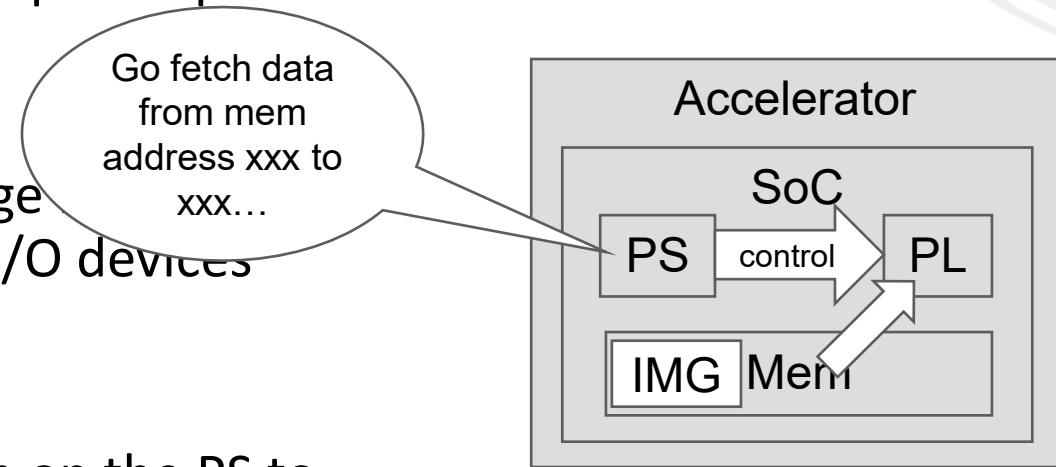
- Implement an accelerator on the PL to speed up computations for the NN inference
- Assume the processor has got the image in the memory--no need to work with other I/O devices such as a camera



What you will do

- **The task:**

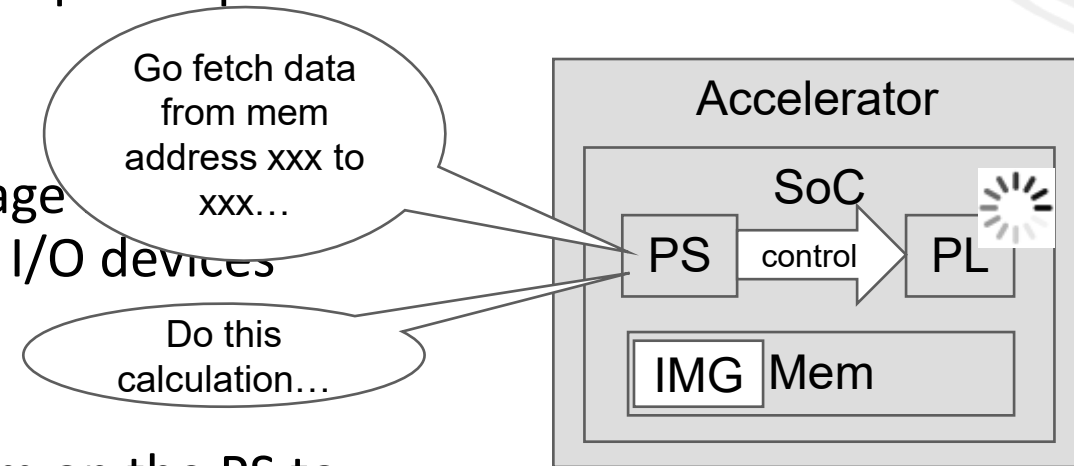
- Implement an accelerator on the PL to speed up computations for the NN inference
- Assume the processor has got the image memory--no need to work with other I/O devices such as a camera
- Modify the st_mnist inference program on the PS to control the PL. For example:



What you will do

- **The task:**

- Implement an accelerator on the PL to speed up computations for the NN inference
- Assume the processor has got the image memory--no need to work with other I/O devices such as a camera
- Modify the st_mnist inference program on the PS to control the PL. For example:



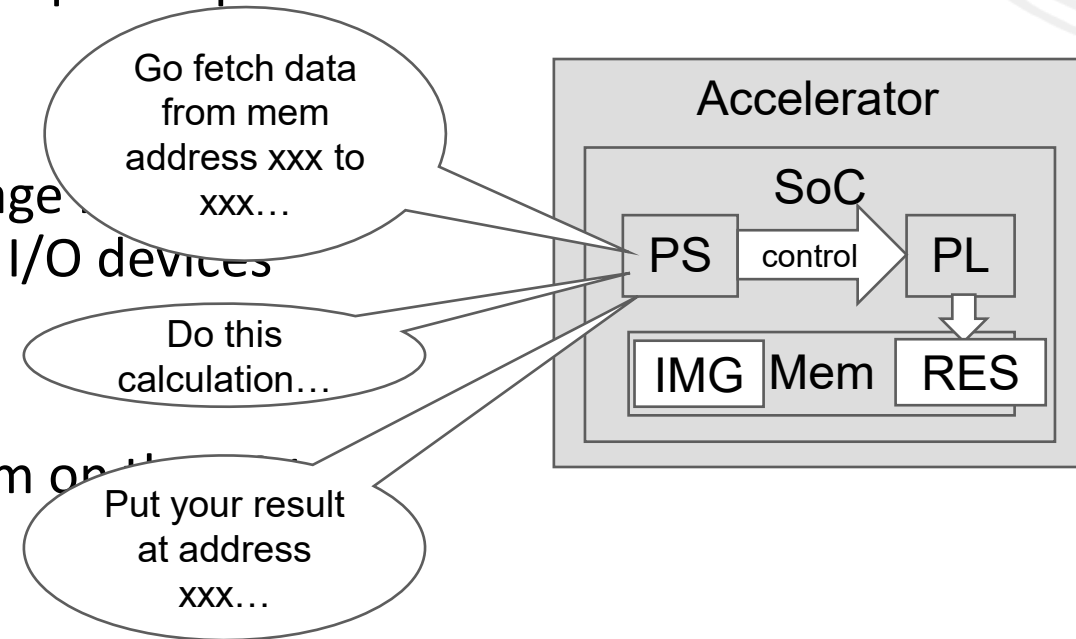
What you will do

- **The task:**

- Implement an accelerator on the PL to speed up computations for the NN inference

- Assume the processor has got the image memory--no need to work with other I/O devices such as a camera

- Modify the st_mnist inference program on the PL to control the PL. For example:



What you will do

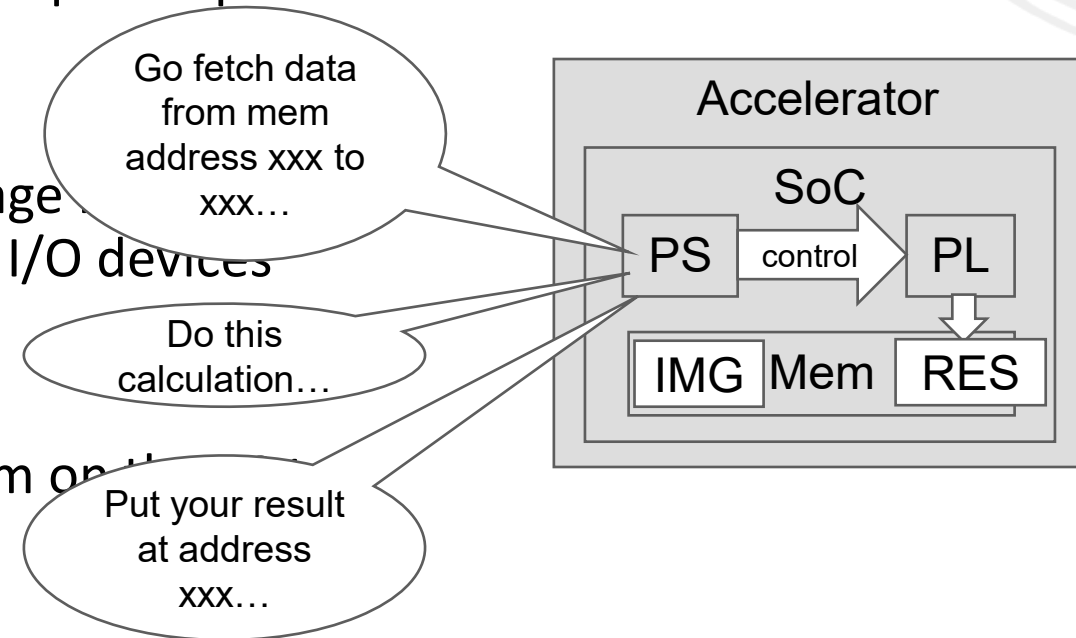
- **The task:**

- Implement an accelerator on the PL to speed up computations for the NN inference

- Assume the processor has got the image memory--no need to work with other I/O devices such as a camera

- Modify the st_mnist inference program on the PL to control the PL. For example:

- Note that you will also need to add corresponding interfaces on the PL to make them work together



Project Schedule

- **Tutorials:**

- In .pdf format, given out on Studium in .pdf format
- Intention: hands-on experience; content probably reusable for your project
- Basics: working with the I/O on the board
- Basics: controlling the I/O from the PL side
- Intermediate: PS-PL data transfer using AXI buses and DMA



Project Schedule

- **Lectures**

- One lecture reviewing the tutorials
- Rest of the lectures: follow-up discussions with each group regarding progress and plan (you must attend at least 2 of the 3 discussions!)

- **Lab hours**

- Computer rooms booked for ASPLoC
- Note: TAs will not be present in the room. But we will be in the office and stay reachable



Grading

- **Prerequisite for passing the course:**
 - Pass all labs
 - Attend at least 2 of the three follow-up discussions with TA
 - Return your boards at the end of the course
- **Evaluated: final presentation**
- **Grade 3:**
 - Deploy the neural network inference program on the board
 - In the final presentation, report your analysis on the program behavior:
 - Time spent on each layer?
 - Time spent on each part of each layer (data preparation, matrix multiplication...)?
 - What is the most time-consuming part of the NN inference, and how can it be accelerated?



Grading

- **Grade 4:**
 - Deploy the neural network inference program on the board
 - Implement an accelerator on the FPGA for critical steps of the program
 - In the final presentation, demonstrate and explain your accelerator design:
 - How does the processor communicate with the programmable logic? (What is done on the FPGA side, and how do you modify the program on the processor side?)
 - What tasks are offloaded onto the accelerator?
 - Note that you are allowed to reuse the code in lab 2



Grading

- **Grade 5:**

- Deploy the neural network inference program on the board
- Implement an accelerator on the FPGA for critical steps of the program
- In the final presentation, report your analysis on the program behavior:
 - (Same as grade 3)
- In the final presentation, demonstrate and explain your accelerator design:
 - (Same as grade 4)
- In the final presentation, analyze your result and reflect on your design:
 - How much speedup do you get for having the accelerator?
 - How much (%) FPGA resources have you used? Have you maxed out its potential?
 - Why did you make each design choice, and what result did you expect to have?
 - Does the final result match your expectation, and why/why not?

